

Implement Linear Regression from Scratch using Gradient Descent

Instructions:

1. Setup Your Environment

- Create a new Python notebook on your local machine (Jupyter, Colab, etc.).
- Make sure you have **Python** installed (version 3.x).
- Import the necessary libraries: `numpy` for computations and `matplotlib` for plotting.

2. Generate Synthetic Data

- Create or import a dataset with a simple linear relationship:
 - Define the number of samples (e.g., 100).
 - Create feature values (x) randomly.
 - Define true parameters (`theta_true`) and generate target values (y) with some noise.
- Visualize the data points using a scatter plot.

3. Define the Hypothesis Function

- Write a function `hypothesis(X, theta)` that calculates predicted values:

$$h_{\theta}(x) = X \times \theta$$

4. Implement the Cost Function

- Implement the Mean Squared Error (MSE) cost function:

$$J(\theta) = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

- This function will help us evaluate how well the model fits the data.

5. Calculate the Gradient

- Derive and implement the gradient of the cost function with respect to each parameter:

$$\nabla_{\theta} J(\theta) = \frac{1}{m} X^T (h_{\theta}(X) - y)$$

6. Perform Gradient Descent

- Initialize parameters (`theta`) with zeros or random values.
- Set the learning rate (`alpha`), e.g., 0.01.
- Iteratively update parameters:

$$\theta := \theta - \alpha \times \nabla_{\theta} J(\theta)$$

- Run for a fixed number of iterations (e.g., 1000).
- Track and plot the cost value over iterations to visualize convergence.

7. Evaluate Your Model

- After training, print the final parameters.
- Plot the fitted line against the original data points.
- Optionally, test your model on new data.